NASA SPACE APPS 2026 | 100% OFFLINE-FIRST | 21/21 SUITES PASSED (124 TESTS)

Sol Check — Complete System Architecture & Team Manual

Autonomous, offline-first health evaluation platform for long-duration crewed Mars surface exploration (Sol 43 Post-Solar Storm) under 20–40 minute round-trip interplanetary latency.

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1. The Deep-Space Operational Challenge

> **The Interplanetary Latency Barrier:** At an average orbital distance of 225 million kilometers from Earth, electromagnetic signals face a 20-minute one-way delay (40 minutes round-trip). In acute clinical anomalies, post-solar particle events, or insidious physiological deconditioning, astronauts cannot wait for Mission Control on Earth. Clinical intelligence must be autonomous, explainable, and air-gapped on-device.

2. System Architecture & Component Hierarchy

Edge Sensors & Video

Local device camera capturing sub-surface blood volume pulse (BVP) via contactless optical rPPG. Zero cloud transmissions.

Local Neural WASM

MediaPipe BlazeFace + LiteRT WebAssembly neural runtimes execute completely inside the client sandbox.

Deterministic Engine

7 clinical decision heuristics, rolling 14-sol median baselines, and MAD spread without ungrounded LLM inference.

3. Explainable Clinical Logic Rules

Rule ID	Rule Name	Clinical Trigger Condition	Operational Action
`entry-quality`	Physiological Bounds	Values outside survivable human limits or SBP ≤ DBP	Rejects corrupted sensor data; orders recalibration
`combined-sleep-mood`	Behavioral Cascade	Sleep < 6.0h AND Mood ≤ 2 across ≥ 3 consecutive sols	Escalates to CMO; orders sleep hygiene & duty rest
`sustained-trend`	Longitudinal Drift	≥ 3 consecutive readings shifting monotonically	Flags insidious deconditioning (e.g. fluid shifts)
`baseline-change`	Acute Anomaly	Single reading deviates by ≥ metric demo delta	Highlights 3D Twin hotspot; emits clinical warning
`reported-symptoms`	Symptom Recurrence	Active reported symptoms (e.g. cephalalgia, fatigue)	Correlates symptoms with active mission hazards
`archive-replay-context`	Radiation Event Nexus	Correlated with Sol 41 Curiosity RAD storm (>15 µGy/h)	Contextualizes crew fatigue with habitat storm shelter
`data-unavailable`	Evidence Disclosure	Fewer than 5 distinct measured sols recorded	Discloses uncertainty; prevents premature triage

4. 3D Astronaut Digital Twin Features

The mission dashboard hosts an interactive 3D digital twin (`astronaut_sol_check.glb`) authored by the team in Blender and rendered locally with Three.js:

- **Pulsating Chest Marker:** Real-time CSS animation frequency dynamically synced to the astronaut's resting heart rate.
- **Anatomical Hotspots:** Head (Mood), Visor (Sleep), Chest (Heart Rate), Torso (Body Temp), Abdomen (Blood Glucose), Limbs (Symptoms).
- **Intelligent Fallback:** Seamlessly switches to an accessible 2D vector figure on hardware without WebGL or under reduced motion settings.

5. ISS Cognitive Performance Laboratory (PVT-B)

Implements the NASA Human Research Program (HRP) **Brief Psychomotor Vigilance Task (PVT-B)** deployed aboard the International Space Station:

- Microsecond precision millisecond timing measuring reaction speed and sustained vigilance.
- Detects lapses (>500ms) and false starts (<100ms) caused by circadian desynchrony and fatigue.
- Calculates NASA ISS Vigilance Scores and prescribes operational countermeasures.

6. Multi-Platform Deployment & Run Instructions

A. Run Web Production Demo (100% Offline)

```
 npm ci
npm run build
npm run preview -- --host 0.0.0.0 --port 4173
```

Open `http://localhost:4173/`. Go to **Demo and data**, confirm **Ready to run offline**, and test with your internet disconnected.

B. Run on iPhone 15 & Android via Expo Go

```
 cd solcheck-expo
npx expo start --go --lan
```

Scan the terminal QR code or open `exp://10.52.237.41:8081` directly in the Expo Go app.

7. Quality Assurance & Verification

> **Verification Suite:** 21 test suites passed, 124 unit and integration tests passing in 511ms. Automated headless Chrome DevTools Protocol (CDP) verification tests verify 100% offline navigation across all 8 mission routes.

8. Team Roster & Attributions

Crew Engineers & Authors: Ahmed Shacker, Ahmed Sherif, Andrew Sameh, David Michael, Hassan Mohsen, Ziad AbdElhaleem.

Coach: Mariam Mohamed | **Local Lead:** Dr. Moatasem Mahran.

3D Asset: Original astronaut 3D model designed and authored by the team in Blender.
